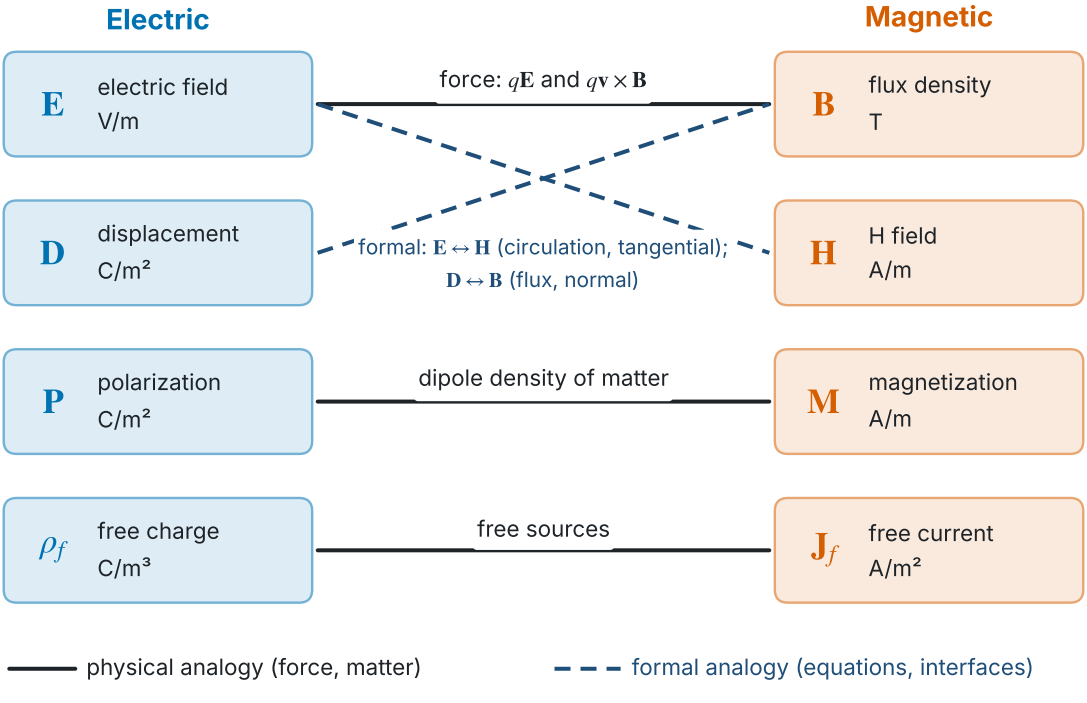
Electric
Magnetic
E
electric field
V/m
force: $q\mathbf{E}$ and $q\mathbf{v}\times\mathbf{B}$
B
flux density
T
D
displacement
C/m²
formal: $\mathbf{E}\leftrightarrow\mathbf{H}$ (circulation, tangential);
$\mathbf{D}\leftrightarrow\mathbf{B}$ (flux, normal)
H
H field
A/m
P
polarization
C/m²
dipole density of matter
M
magnetization
A/m
ρ_f
free charge
C/m³
free sources
$\mathbf{J}_f$
free current
A/m²
physical analogy (force, matter)
formal analogy (equations, interfaces)